BODY—CORROSION AT BOTTOM OF DOOR OR TAILGATE AT HEM FLANGE—REPAIR PROCEDURES	Article No. 01-9-8

FORD: 1997-2001 F-150, F-250 LD

ISSUE

Some vehicles may exhibit corrosion at the hem flange at the bottom of any door/tailgate where the outer panel is folded over the inner panel. This may be caused by the adhesive absorbing water during the production process.

ACTION

Repair corrosion at hem flange using new design special hem flange tools and procedures. Refer to the following Service Procedure for details.

SERVICE PROCEDURE

NOTE

READ THIS REPAIR PROCEDURE COMPLETELY BEFORE PERFORMING ANY SERVICE ACTIONS.

Inspect the inside of the door/tailgate at the lower edge only to determine which service procedure to perform:

- Procedure “A”: Corrosion/Bubbling at the Door Bottom Hem Flange
- Procedure “B”: Corrosion/Bubbling at the Tailgate Hem Flange

This repair procedure can be used on third door, fourth door, and tailgates of trucks.

Refer to the materials listing later in the TSB for materials required for each procedure.

NOTE

THE USE OF FORD OR MOTORCRAFT BRAND PRODUCTS ARE MANDATORY ON ALL REPAIR CLAIMS PAID BY FORD UNLESS OTHERWISE SPECIFIED IN EMISSIONS, SAFETY RECALLS, OWNER NOTIFICATION PROGRAMS, TSBs OR OTHER COMPANY PUBLICATIONS. THE COST OF NON-FORD/MOTORCRAFT PRODUCTS USED FOR A WARRANTY OR ESP/ESC, OR AWA REPAIR WITHOUT JUSTIFYING THEIR USE (I.E., EMERGENCY REPAIR), IS NOT REIMBURSABLE.

Examples of Ford products are: adhesives, sealers, solvents, cleaners, washes, and other special products listed in the Ford Car Care Products Manual.

Technicians should wear the appropriate eye, hand and hearing protection while performing these repair procedures. Adhesives and sealers used in these procedures should be used in a well-ventilated area.

PROCEDURE “A”: CORROSION/BUBBLING AT THE DOOR BOTTOM HEM FLANGE

1. Follow the appropriate Workshop Manual procedure for the removal of the interior door panel.
2. Use shop air to blow out the drain holes in the bottom of the door and apply 25mm (1″) masking tape to the outer door skin at the bottom to prevent scuffing.
3. Use a hammer and screwdriver to slightly lift the corner of the hem flange.
4. Insert TOOL 501-078/1(RT) or TOOL 501-078/2(LT) in an air impact gun (620 kPa (90 psi) air pressure is required for consistent tool operation). Using the special tool, lift the hem flange continuing across the bottom of the door and stop just before the end.
5. Use an angle grinder or wire brush to remove all corrosion on the bottom inside hem flange and inner door panel. Follow by blowing off with compressed air.
6. Using appropriate hand protection, brush on Corrosion Converter (Sem 69508) on the corrosion area. Use light coat and let flash dry 2-5 minutes. Follow by using a damp cloth to wipe off corrosion residue.
7. Sand with DA sander using 360 grit sandpaper at the lower areas of the door where paint was disturbed, keeping the repair area small. Featheredge for a paint blend as necessary. Blow off the panel with clean compressed air. Clean the panels with wax and grease remover.
8. Mix and brush on Ford-approved epoxy or self-etching primer to inside hem flange areas on the bottom of the door. Bake at 60°C (140°F) panel temperature per manufacturer’s label instructions.

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9. Install the Motorcraft TA-1 adhesive cartridge into the application gun (501-040). Squeeze out some material to purge the cartridge. Install the tip on the cartridge and squeeze out some material to make sure of even mixing of the adhesive components.
10. Apply the adhesive between the hem flange and bottom of the inner door panel. Work time is 45 minutes at 21°C (70°F). Adhesive cure time is 8 hours.
11. Use TOOL 501-080 to re-hem (close down) the hem flange. Start in a corner and insert the center of the tool over the area to close down, then hit with a hammer.
12. Install the Motorcraft TA-2 adhesive cartridge into the application gun (501-040). Squeeze out some material to purge the cartridge. Install the tip on the cartridge and squeeze out some material to make sure of even mixing of the adhesive components.
13. Apply Motorcraft TA-2 Urethane Seam Sealer around the door hem flange to give a finished appearance.
14. Remove any residual sealer with adhesive remover. Sealer is paintable in 10 minutes and within 24 hours.
15. Mask the door for color application.
16. Mix and apply Ford-approved basecoat color to the inside of the door area and allow material to flash off.
17. Mix and apply Ford-approved clearcoat to inside of the door and bake at 60°C (140°F) panel temperature per manufacturer's recommended baking time.
18. Spray SuperSeal (TA-8) aerosol rust preventive to the inside of the door at the hem location. Do not plug drain holes.
19. Demask and reinstall the inner door panel.

PROCEDURE "B": CORROSION/BUBBLING AT THE TAILGATE HEM FLANGE

1. Follow the appropriate Workshop Manual procedure for the removal of the tailgate and tailgate mouldings as necessary.
2. Use shop air to blow out the drain holes in the bottom of the tailgate and apply 25mm (1″) masking tape to the outer tailgate skin at the bottom to prevent scuffing.
3. Use a hammer and screwdriver to slightly lift the corner of the hem flange.
4. (Depending on preference) Insert TOOL 501-078/1(RT) or TOOL 501-078/2(LT) in an air impact gun (620 kPa (90 psi) air pressure is required for consistent tool operation). Using special tool, lift the hem flange continuing across the bottom of the tailgate and stop just before the end.
5. Use an angle grinder or wire brush to remove all corrosion on the bottom inside hem flange and inner tailgate panel. Follow by blowing off with compressed air.
6. Using appropriate hand protection, brush on Corrosion Converter (Sem 69508) on the corrosion area. Use light coat and let flash dry 2-5 minutes. Follow by using a damp cloth to wipe off corrosion residue.
7. Sand with DA sander using 360 grit sandpaper at the lower areas of the tailgate where paint was disturbed, keeping the repair area small. Featheredge for a paint blend as necessary. Blow off the panel with clean compressed air. Clean the panels with wax and grease remover.
8. Mix and brush on Ford-approved epoxy or self-etching primer inside hem flange areas on the bottom of the tailgate. Bake at 60°C (140°F) panel temperature per manufacturer's label instructions.
9. Install the Motorcraft TA-1 adhesive cartridge into the application gun (501-040). Squeeze out some material to purge the cartridge. Install the tip on the cartridge and squeeze out some material to make sure of even mixing of the adhesive components.
10. Apply the adhesive between the hem flange and bottom of the inner tailgate panel. Work time is 45 minutes at 21°C (70°F). Adhesive cure time is 8 hours.
11. Use TOOL 501-080 to re-hem (close down) the hem flange. Start in a corner and insert the center of the tool over the area to close down, then hit with a hammer.

12. Install the Motorcraft TA-2 adhesive cartridge into the application gun (501-040). Squeeze out some material to purge the cartridge. Install the tip on the cartridge and squeeze out some material to make sure of even mixing of the adhesive components.
13. Apply Motorcraft TA-2 Urethane Seam Sealer around the tailgate hem flange to give a finished appearance.
14. Remove any residual sealer with adhesive remover. Sealer is paintable in 10 minutes and within 24 hours.
15. Mask the tailgate for color application.
16. Mix and apply Ford-approved basecoat color to the inside of the tailgate area and allow material to flash off.
17. Mix and apply Ford-approved clearcoat to inside of the tailgate and bake at 60°C (140°F) panel temperature per manufacturer's recommended baking time.
18. Spray SuperSeal (TA-8) aerosol rust preventive in the inside of the tailgate at the hem location. The inside of the tailgate can be accessed through the drain holes in the bottom of the tailgate using an extension nozzle on the SuperSeal spray can. Ensure drain holes are not plugged after applying.
19. Demask and reinstall the tailgate.

PROCEDURES "A" AND "B" MATERIALS LIST

PART	PART NUMBER
Motorcraft Metal Bonding Adhesive	TA-1
Motorcraft Urethane Seam Sealer	TA-2
SuperSeal	TA-8
Stone Abrasion Material (obtain locally)	N/A
Corrosion Converter (Sem #69508) (obtain locally)	N/A

PROCEDURES "A" AND "B" TOOLS

SPECIAL TOOL
501-078/1 RH Hem Opening Tool (Tools are direct ship from O.T.C.)
501-078/2 LH Hem Opening Tool
501-080 Hem Closing Tool
501-040 Adhesive Gun (Adhesive Gun not included in kits. This is a commonly used commercially available gun. Must be ordered separately from Rotunda.)

LABOR CLAIMING EXAMPLE

2000 F-150 4X4 Super Cab - Dealer is repairing door flange on one (1) front door and one (1) rear door. Dealer would claim operations as follows:

- 010908AA (0.9) - Time to repair front door
- 010908BA (0.7) - Time to repair rear door
- 010908P101 (0.5) - Basic paint operation (once per visit)
- 010908AAP (0.5) - Time to paint front door
- 010908BAP (0.5) - Time to paint rear door

Material allowance is calculated only on the Paint labor time (010908P101, 010908AAP and 010908BAP). Total the paint labor time, multiply time x $14.75 for the total material allowance. Use (PAINT) as the causal part for claiming material allowance.

- Part #: "Paint"
- Qty.: 1.5
- Ea. Amt.: $14.75
- Total = $22.12

This repair would pay 3.1 hours (0.9 + 0.7 + 0.5 + 0.5 + 0.5 = 3.1) for the total job plus 1.5 hours (0.5 + 0.5 + 0.5 = 1.5) for the material allowance.

PART NUMBER	PART NAME
TA-1	Motorcraft Metal Bonding Adhesive
TA-2	Motorcraft Urethane Seam Sealer
TA-8	SuperSeal
Obtain Locally	Stone Abrasion Material
Obtain Locally	Corrosion Converter (Sem #69508)

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: Eligible Under The Provisions Of Bumper To Bumper Warranty Coverage

OPERATION	DESCRIPTION	TIME
010908AAP	Additional Time To Paint For Operation AA	0.5 Hr.
010908AA	Inspect And Perform Hem Flange Corrosion Procedure - Includes Time To Mask Door Opening (One Front Door)	0.9 Hr.
010908ABP	Additional Time To Paint For Operation AB	1.0 Hr.
010908AB	Inspect And Perform Hem Flange Corrosion Procedure - Includes Time To Mask Door Opening (Two Front Doors)	1.6 Hrs.

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010908BAP	Additional Time To Paint For Operation BA	0.5 Hr.
010908BA	Inspect And Perform Hem Flange Corrosion Procedure - Includes Time To Mask Door Opening (One Rear Super Cab Door)	0.7 Hr.
010908BBP	Additional Time To Paint For Operation BB	1.0 Hr.
010908BB	Inspect And Perform Hem Flange Corrosion Procedure - Includes Time To Mask Door Opening (Two Rear Super Cab Doors)	1.2 Hrs.
010908CAP	Additional Time To Paint For Operation CA	0.5 Hr.
010908CA	Inspect And Perform Hem Flange Corrosion Procedure - Includes Time To Mask Door Opening (One Rear Super Crew Door)	0.9 Hr.
010908CBP	Additional Time To Paint For Operation CB	1.0 Hr.
010908CB	Inspect And Perform Hem Flange Corrosion Procedure - Includes Time To Mask Door Opening (Two Rear Super Crew Doors)	1.6 Hrs.
010908DAP	Additional Time To Paint For Operation DA	0.5 Hr.
010908DA	Inspect And Perform Hem Flange Corrosion Procedure Tailgate	0.6 Hr.
010908P101	Can Be Claimed Once Per Visit Even If Multiple Panels Are Involved	0.5 Hr.

DEALER CODING

BASIC PART NO.	CONDITION CODE
1520124 RT. FT. DOOR	D7
1520125 LF. FT. DOOR	D7
1624630 RT. REAR SUPER CREW DOOR	D7
1624631 LF. REAR SUPER CREW DOOR	D7
1824630 RT. REAR SUPER CAB DOOR	D7
1824631 LF. REAR SUPER CAB DOOR	D7
8340700 FLAIRSIDE TAILGATE	D7
8340700 SUPER CREW TAILGATE	D7
9940700 STYLESIDE TAILGATE	D7

OASIS CODES: 111000